

One evening, once. Touch the appendix – if it is warm, you have a few hours. If it is cold, you have next year.

THE CYCLE – IT IS ALL ONE NIGHT

1	Day 1, late afternoon into evening. Spathe unrolls. The appendix goes hot – 10 to 20 °C above the air, unmistakable to a finger – and the plant stinks. Stigmas receptive. This is the entire window.
2	Day 1, late evening. Anthers open, sometimes by 10 pm. Pollen falls into the chamber base all night. Too late to apply, too early to collect.
3	Day 2, morning. The spadix wilts and becomes climbable; trapped insects walk out loaded. Collect pollen now.
4	Day 2 onward. No scent, no heat, nothing attracted – and the spathe still looks perfect. That is the trap for you, not the flies.

WHAT YOU NEED

- A finger – the heat is the test
 - Soft brush
 - Small airtight tube
 - A fridge, or a cool dark cupboard
- Scalpel, to cut a window in the spathe
 - Organza bag, if the cross matters
 - Torch. You will be out at dusk

READING THE INFLORESCENCE

Appendix warm , strong smell, evening	Receptive. Pollinate now. There is no second chance on this spathe.
Spathe open, appendix cool, no smell	Yesterday's flower. Female phase is over.
Pollen in the chamber base, next morning	Male phase. Collect – do not apply here.
Flies moving inside	Working as designed. Leave them; they go at dawn.
Spathe still standing days later	Means nothing. The spathe outlasts the function.
Tip of the appendix chewed	It will abort. Every appendix-eaten inflorescence in one study did, even ones already setting fruit.

THE METHOD

- 01 **Go out at dusk, not dawn.** Every other card in this set says the opposite. *Arum* is an evening plant, and at least one of its pollinating flies is nocturnal with an endogenous clock.
- 02 **Test by touch.** Warm appendix and a strong faecal or urinous smell. Cold appendix means you are a day late, whatever the spathe looks like.
- 03 **Collect on the morning of day 2** from a different inflorescence, where pollen has fallen into the chamber base.
- 04 **Cut a small window** in the chamber wall and brush pollen directly onto the stigmas.
- 05 **Bag it, and re-bag immediately after.** The study that re-bagged got 80–100 % set; one that does not mention re-bagging got 15 %.
- 06 **Label it.** Roughly three months to a ripe berry.

Sources: Knoll 1926; Diaz et al. 2006; Gibernau et al. 2003; Albre et al. 2003; Gibernau, Macquart & Przetak 2004; Leclerc et al. 2025; Sowter 1949.
Full guide, with the evidence behind every line: aroidpedia.com/arum-pollination

Everything on this side contradicts something you will read elsewhere. Each one changes what you actually do.

1 • ONE CLONE IS ENOUGH

Bagged and left alone, 100 inflorescences across three taxa set no seed at all. That result is reported everywhere as self-incompatibility. **It is not.**

In the same study, pollen carried by hand from **another inflorescence of the same clone** gave a **100 % infructescence set**, with seeds equal in number and weight to outcrossed ones. Self pollen germinates on the stigma as well as outcross pollen does.

The barrier is **timing, not compatibility**. One spathe cannot pollinate itself — but a plant's inflorescences open days apart, and stigmas stay receptive from about 12 hours before anthesis to 12 hours after. **So a single plant is workable. Carry the pollen.**

2 • IT IS NOT A FISH-TRAP

The bristles are **not** a one-way valve, and the insects do not walk in. They **slip off the inside of the hood and fall**, because it is coated in a fatty oil over downward-pointing papillae.

Knoll proved it in 1926 by cutting a chamber free and **turning it upside down**: the same insects then climbed the same surface perfectly well. And they escape **up the spadix**, which wilts overnight and becomes climbable — not past the bristles, which stay put.

Why you care: nothing you do to the bristles matters, and a spathe that has been opened or damaged has lost its slide surface, not its valve.

3 • THE HEAT IS NOT A LURE

A heated model inflorescence, run on **thirteen separate days**, caught nothing — while a scent-baited model beside it caught the normal fauna. Offered warm tubes, the main pollinator showed **heat avoidance**.

The heat volatilises the scent. **For you it is a thermometer, not a signal** — which is exactly what makes it the most reliable test in the genus.

POLLEN — CHILLING ROUGHLY DOUBLES IT

Left in the spathe, outdoors	2 days, then falling fast
Dark dry cupboard, 15 °C	4 days , weakly viable to 7
Refrigerated, 8 °C	72 hours , then rapidly to zero
Dry at room temperature	Under 48 hours in one species

SEED

Sow the year you harvest. Seed kept to a second season gave **zero** germination; fresh seed from the same source gave 29 %.

Sow outdoors and let it freeze. At room temperature in daylight, nothing germinated at all; in darkness, 20 %; **frozen outdoors through the winter, 86–92 %**.

Then wait. **Nothing appears above ground for two seasons**, the adult leaf comes in the fourth, and a seed-grown plant **rarely flowers before the seventh year**. Division is the sensible route; seed is for making something new.